

# Alvaro Muñoz NET Serialization Detecting and defending vulnerable endpoints

**Alvaro Muñoz NET Serialization Detecting and defending vulnerable endpoints** - All Hacking Cons, YouTube.

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## Content

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# Alvaro Muñoz NET Serialization Detecting and defending vulnerable endpoints

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## Description

LocoMocoSec 2018 Kona Hacking conference #hacking, #hackers, #infosec, #opsec, #IT, #security

## Transcript

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thank you um can you hear me yeah so all right um thank you Jim for the introduction um so as Jim said I'm Alvar Munoz also known as pontester in a social network and Twitter GitHub and so on I work as a principal security researcher with fortify team I mainly do a static analysis related research and probably will I focus on web application security right

uh why I find more interesting is getting remote code execution on on web applications and I have responsibly reported a critical remotely execution vulnerabilities to companies like Microsoft Oracle work based Salesforce and so on um so one of them in Microsoft I will be sharing with you today hopefully if we have time so apart from that I have some experience around digitalization I started not in 2015 when as Jim said but actually in 2012 playing with XML decoder and access stream trying to get remote code execution in restful web services right so we were able to get remote code execution and some libraries like restulate and rest easy and then I moved into the binary format of java digitalization right and I found uh to what I to my knowledge is the first remote code execution gadget for Java digitalization but for the binary format back in 2013 and it was not in Apache Commons collection but actually in the spring libraries right so unfortunately uh this uh Gadget didn't become as popular as the Apache Commerce collection Gadget became and I'm saying unfortunately because uh we all saw that this was like a Tipping Point right and when um Christopher hope and Gabriel Lawrence found this gadget the community started looking for these vulnerabilities they started fixing them and so on so we could have saved ourselves like a couple of years by starting back in 2013 or even even before so at this point in 2015 Christopher huff and gabrielores published the Apache common collection Gadget and everything changes right all the researchers start looking into Java digitalization they start publishing vulnerabilities like in weblogic web sphere and many other applications so I took my interest in Java digitalization I I found like a technique to bypass what is known as the look ahead object digitalization that is like the standard mitigation technique for this kind of attacks in Java and also I found like multiple gadgets are leading to remote code execution including one in the Java runtime itself so not requiring any third-party libraries in order to work so then I switch gears and I started looking into Json to close the circle right so from XML to Binary and then to Json and I publish some approaches to get remotely execution in Json these serializers back in last year in in black hat and Defcon so most of this is Java right and this is what is known as the Java digitalization apocalypse and I think well it's bad because I mean there was many applications that were affected by these bugs but in the other hand it was good because we got this categorial together to go into the OAS top 10 right and we as a security Community we have knowledge about the latest trends and latest categories like I don't know SPL injection or jndi injection but developers don't normally have this visibility in what vulnerabilities they should focus on and I think that having a secure digitalization in the OS top 10 is a good thing in order for developers to start looking and fixing these vulnerabilities so at that point I decided to take a look into .net because well I'm not a .net developer I'm not a .net researcher but since I had some previous experience with digitalization I decided to take a look because apparently there was a previous research around .net digitalization but there was no remotely execution got it and has happened with Java before it was not until a remote execution Gadget was published that the community started fixing and taking care of this vulnerability so I decided to look for one in order to start you know the ball moving so what I'm going to present today is the result of This research so I will first introduce a brief introduction into serialization then we will be reviewing what are the most common uh don't net serializers and formatters and then we will move to the more like Builder and Defender part of the top that is finding

vulnerabilities and fixing them all right so um Marceline pickles I will be using Rick and Morty picture because why not so this is Rick all right he's an instance an object living in the uh VM and at some point the developer want to take Rick out of the VM right so he wants to put it in that file system or into the database or send it across the network so he put a rig across the magic portal and then he gets a pickle Rick all right so at a later stage we want to recover the object back in the VM so we will put pickle Rick back into the magic portal and then recover and reconstruct rig again all right so there is one thing in this slide that is critical for this digitalization attacks to work so anyone something that is like the design flow of all these formats like Dev so say that again sorry yeah well this is just a very simplification a simplified model of a digitalization but the issue I was referring to is that we're not sending just the the pickle itself but we are sending a pickle Rick as long as with the satellite data as well so um this is what is known as the type discriminator and basically if you are familiar with Java serialization uh even though it's a binary format you will find like class names or type names in the in the serialized data same thing with dotnet for example this is json.net and you will find some dollar attributes dollar type attributes containing something that looks like um dotnet type and Dot net assembly names so if the attacker can control the type discriminator and they change speaker rig with a pickle multi then the server will blindly trust that and then we'll instantiate an instance of more instead of the of the brick one right so there is probably nothing really dangerous here because they are going to invoke the default Constructor of this multi-class which is uh it takes no parameters so there's nothing where the attacker can modify the behavior of this and and just get arbitrary called execution however apart from the default Constructor there will be other methods that will get invoked like for example in Java we'll get the read object and read result methods and in.net the same thing we have like this additionalization Constructor over here and this realization callbacks even some annotations that we can use to annotate methods that are going to be invoked by the framework in order to fully reconstruct the object right same thing with XML and Json um the these libraries will normally just invoke the default Constructor and then invoke Setters in order to populate the data in the class types and we they will be invoking centers in order to do that all right so all that the attackers need to do is find the gadget right and I got yet is nothing else that a type or a class that is sitting in the Target application class path so they cannot provide implementation for the Prototype they can only provide the type name and since they can control the values for these properties of this class then they are able to actually modify the behavior of the methods that are going to involve to be invoked like for example the visualization callbacks or the setters so this is more like a builder and Defender talk and I'm not going to focus on how to find and and yeah well find gadgets leading to remote execution but I think it's good to show you an example of a very simple one leading to remote execution so you can have like an idea of how they look like so this is a system Windows data object data provider is a type in the windows guide so it's available without any third-party dependencies required and it contains three different Setters right that will initiate some invocations that will end up invoking this method over here and here we are basically just using reflection to invoke method name on an object instance passing some parameters right and those are exactly the ones that we can control uh see if we can control the values that are passed to the Setters right so if an

attacker provides a piece of json.net data like this one to a vulnerable endpoint the vulnerable endpoint will basically instantiate the object data provider and then it will invoke the Setters for the object instance for the method parameters and for the method name and as we saw before this will initiate this uh this calls here and we'll end up invoking the reflection and getting the um in this case the calculator to pop up right so I'm not going to talk more about gadgets if you want to check which Gadgets are available for Java I visit the yso serial project if you want to know more about gadgets in .net I grow this project that is called ysocial.net that basically it can be used to generate payloads and gadgets for different formats in the .net family so this is a simple example for example just invoke it like the one that we just saw if you want to attack the lesson on that just specify the formatter specify the gadget specify the command and it will generate the payload for you to use in your penetration test and pocs and so on so with that let's move into reviewing what formatters and serializers are vulnerable in the donut family so as I said this is nothing new and back in 2012 uh James Forso from the Google project zero was already presenting on how passing and trusted data to a binary formatter or net data contract serializers could lead into some malicious code being run at that time there was no remote of execution guide yet and I think that he only presented a gadget that was able to delete files from the file system or initiate some SMB relay attacks right so as happened with Java before since there was since there were no remote execution guide yet no one paid really um you know attention to these vulnerabilities and there were like many many vulnerabilities uh related with net visualization until until last year probably so last year in 2017 my colleague Alexander Middles and myself we found that remote execution got that in .net and that was followed by James Forso presenting two more so there are like a number of them right now so um as we said uh James presented a couple of formatters that are vulnerable in their default configuration those are binary formatter and net data contract serializer but they are more right these are the ones like I said that are vulnerable in their default configuration they should never be used with untrusted data at least in this default configuration and they include the binary formatter one that is the one that James presented but this one is also used internally by other serializers right for example this binary message format that is used by the Microsoft message queue system so if you can send them arbitrary types you will be able to get arbitrary code execution same thing with the object state formatter that is used by the asp.net view state so if you can change the view State and survive the hmac verification that will also lead to remote execution and we will see more about this later so then we have more like sub formatters using web services 5.json Street Json those are like third-party libraries so the most popular and used one is the binary formatter and this is probably the most common anti-pattern that I found I literally found like thousands of applications in sodan vulnerable to remote execution because they were using exactly this pattern and this is basically well this is very simple uh they are basically instantiating the binary formatter and then serializing some data base64 encoding that data and putting that into a cookie so if that cookie comes back the server is going to be serialized that and since there is no hmac verification it will go in and run arbitrary commands using this gadgets so the first time that I found that was in app Harbor anyone knows up Harbor yeah I know you use it so uh well browsing aside I realized that they were setting this cookie with this a a e a a d that is nothing else that the magic

number for the binary formatter and then base64 encoded so that means that this uh site that is basically like Heroku for .NET applications like to host .NET Applications I was born about remote code execution we reported that and they fix it but then we realized that they even had a blog post explaining how to do exactly what they were doing so instead of keeping some objects in the server side just save some memory and put them in the client side by serializing them and putting them into a cookie so we tell them about this blog post and they now deleted these these blog post however this same advice is everywhere um you can find like multiple gifts and blog posts and even in Microsoft ASP.NET MVC framework was using the very same approach that they silently removed back in 2013 right and this is related with what Jeremy was talking before like if they don't publish like a CV you don't know that there is something that you should update and so on look this is kind of important so let's see a demo this is a vulnerability that I found in Azure um yeah so this is from last year so as you can see even though this research by Game Source of West published in 2012 there were many um developers even Microsoft developers using binary format there with untrusted data so if you're familiar with Azure Active Directory um or application proxy you basically have your intranet here and you have your web application that you want to share right to to expose to your remote employees so your employees will basically connect to the application proxy then will be redirected to the Azure Active Directory well they will be like ask for the credentials that will be checked against your own instance of the Active Directory and then they will be returning a token and then authenticating against the application proxy I think that is very better explained with a video so this is uh we're visiting a application that I'm exposing through this uh Azure application proxy and I'm redirected to the Microsoft single sign-on right I don't even need to log in into Azure because this is a pre-authentication of remote code execution so I will basically send a request through Burp or any proxy and well basically I will just send this request to the repeater so we can see what is going on and as you can see here we get a 302 redirection to the singles and on page in Microsoft and then we get this cookie assigned here right if you recognize this magic number the same one in app Harbor this basically means remote execution so we just uh get what we have to do now is generate aware our payload and we will be basically um this was before I wrote the why.society.net so I had to do it by hand but basically I will be doing a DNS resolution against my own DNS server and I will be linking the computer name just to prove that the remote execution was not in my own intranet servers but actually in Azure servers so with that I will just generate the payload during the process of generation of generating the payload the payload will get executed so I will get a connection back to my DNS server from my own VM as you can see here so that's my VM computer name so nothing interesting so far but now if we copy paste the payload generated by our card yet or payload generator into the cookie and we send this request to Microsoft Word to Azure Network we will see that we will get some connection back from a different servers actually not just one so for example the first one is from this server down here that is Microsoft servers and then if I submit the request again I will get a different connection which means that I can compromise all the application proxies that are working like in a load balanced way which means that I can intercept any traffic going to any intranet of customers using this Azure thing so that was pretty bad the good thing is that Microsoft only had to patch their own system and

that was quite easy and they did it very quickly all right so apart from those um these serializers that are vulnerable in their default configuration there are other serializers that are known as template serializers that work with an expected type right so if you want to deserialize something they will only work with one type right for example if you're expecting a shopping cart instance then you cannot send like an object data provider payload because that won't work right so those examples examples of that is for example the data contract serializer the data contract json serializer and the XML serializer so in these cases they are still vulnerable but the attacker needs to control not just the data that is going to be deserialized but also the suspected type you may think that this is not very common but I found that for example one good example in dot net nuke that is the most popular content management system in .net and basically at some point they were taking the contents of a cookie that was not signed and that was like an XML cookie I don't know why but anyway they were extracting some type name from this XML and then they were invoking the Constructor of the XML serializer with this data that can be controlled by the attacker so now the attacker can control the deserialized type and then it will be extracting the rest of the XML and then deserializing that so now that I can control the expected type and the data being deserialized which means that the attacker can control the whole object graph and run an arbitrary code in the server so that was reported and was fixed okay so the third group is those serializers that are vulnerable in some specific configuration right so those are for example JavaScript serializer that is the native Microsoft json deserializer and json.net that is the third party one that is even more popular than the Microsoft One so let's see the first one this is JavaScript serializer JavaScript serializer is safe to be used with untrusted data as long as you don't do this that is basically passing a type resolver if you use a type resolver that means that it's going to include the type information along with the json data and then it's going to process that type information so it's going to allow the attackers to instantiate arbitrary types and invoke these setters that will lead to arbitrary code execution same thing with json.net uh but instead of an argument to the constructor in this case it's a property right and they are set in this type name handling which means that it will include and process the type information in the json data so you have to set known because if you set objects arrays or Auto then the framework will be including this information and it will make the endpoint vulnerable all right so how about finding if your applications are vulnerable so they are like different approaches they are the first one probably the most simple one is to inspect the traffic flowing in and out of your applications and look for magic numbers or these signatures of the traffic being serialized data as we saw before this AAE aad is an example of this this is the base64 format for the magic number of the binary formatter but there are other magic numbers like for example for the object State formatter for the XML serializer and they are other encodings so I wrote this simple better plugin that you can freely use and basically we'll use uh we'll check for this magic magic numbers in different encodings so um well I didn't mean to be like a perfect very accurate plugin so it will have some false positives like for example if this magic number is found in some images that will be reported and also it might find that you are sending serialized data but it's not checking if there is a signature for that serialized data so maybe you cannot explode it it's not a false positive because there is serialized data but it's not

exploitable so we will see more about the view State later so the second approach is an active approach where you are basically sending a payload and watching if that payload gets executed and trying to recognize out of bound if that gets executed so example of this is like for example making the server and sleep for a number of seconds and then check if the response gets delayed or for example like I did with Azure you can run a DNS lookup in order to if you control the VNS server to check if the payload is being executed is another approach I don't know if it's passive active or something in the middle but I consider it active because you can you have to actually instrument your virtual machines right so you instrument the virtual machines you monitor what is going on inside of the virtual machine and then if some untrusted data gets into this decentralized methods then we will report that that's good for pre-production but not for production anyway if you have access to the source code that's probably the easiest approach just check if someone tainted data some untrusted data flows into one of these digitalized things if you do that you have to consider that is not a simple well for some formatters like binary formatted it's very simple first for some other formatters such as XML serializer you have to correlate different data flows and also taking control flow into account because there can be a serialization binder that we will see later so you have to consider that but this is something that most of the static analysis tool can do so if you use fortify I know it worked because I grow the support for that so you are safe um also uh fixing vulnerable endpoints right so now you find that your you found that your application is vulnerable sorry so how can you fix it right so the first approach is do you really need to serialize that data and this seems silly but I found like many applications that were using serialization for Strings right and one good case for for this and one good example is the Nancy web application framework the Nancy is basically a web application framework that is very similar to Sinatra framework in in Ruby and they care about security and they implemented C self-protection but instead of putting a unique token into the Cesar cookie they were satellizing that token into binary format serialized data and then putting that into the cookie so making the whole framework and then all the applications built using this framework Boon and about remotely execution because of this implementation of the csoft protection uh same thing with uh type discriminators um Json and XML is okay to to use but you should not use it we use them with type information right so that was the case of the breeze framework Breeze is a dotnet entity management framework they basically suppose it's very similar to Spring data rest so they expose some database entities through our restful web service automatically and they were sending the type information to the JavaScript clients so that made no sense because JavaScript is not going to understand the .net type so they were not using that so because of this a mistake the whole framework was vulnerable to promote the execution and they fix it very quickly in a matter of hours by just setting this time name handling setting to known right so this is actually the default value so probably at some point they were playing with that and they changed that and they remain in the code so I think it's a good idea to explicitly set this secure values explicitly and then even other comments like do not change this please so um okay what what if you really need to serialize data so the first case was when you were serialized in a stream so that was not required now if you have some object graph that you want to persist in the client side for example for performance reasons or whatever reasons then if you are not expecting the

client to modify that data then sign it and verify that the data hasn't been modified right if you do that always use an hmac and never use uh something like md5 or Xiao one with secret and data because that's vulnerable to has lengths extension attacks and this is the way that app Harbor and Azure actually fix their issues and they just sign the the data that they were not expecting anyone to change and then they verify that that change didn't get modified same thing with asp.net if you remember back in 2013 they removed this cookie them data provider and they replaced it with something else and that was basically the same approach of serializing server-side data into the client side but this time they were using the data protector API right they were doing something very similar to this called the data protector API protected bytes and that will basically sign and encrypt the data using the web config validation and encryption keys and then this data won't be able to be tampered with without the server side noticing so another good example of signing and verifying the data is the asp.net View State right so for those of you that are not familiar with the view State it's like a representation of the state of a page so the server is going to send you the state of the page to you as a hidden field for form field and then when you submit the form back you are submitting the previous state so the server can reconstruct the previous state and then continue from there so this is used by asp.net and I think that in the beginning they were not even forcing or they were not signing and encrypting The View state so the problem back in these days was that if there was some sensitive data for example in the view State then the clients were going to be able to extract and leak that data also if the client was able to modify the data because it would not it was not signed and then they could change the application logic so that was back in in the good days like many years ago and at some point in version 4.5.2 Microsoft even started ignoring the enabled view State Mac meaning that even if you set that to false they will always encrypt and sign your view state which is good and they even back ported that back to 1.1 so I was curious and I ran Australian query and I found that more than 200 servers is still running vulnerable versions that are not using any signature or encryption in the view state which makes like a good cryptocurrency botnet or something like that so anyway at some point they improved the cryptographic posture of the view state by actually using a key database on function and this was a very interesting approach because uh if the attacker is able to leak the validation and encryption key then they will be able to actually change the encryption and change uh well not change the encryption but changing the view State and then putting the payload there and get arbitrary code execution so what they did is that they generate some per request a unique strings that they call purposes and what they do is that they run this through a key DF a key derivation function in order to generate some keys that are unique for each request now if the attacker is able to leak this encryption and validation Keys those keys won't be able to Tamper and modify the view State because the ones that will be used are these ones over here right so I just you know connected a debugger and took a look and I visit the account register page and then I check the values for this purposes strings so the first one is primary purpose right and this one is always going to be the same is this string over here with the client state if it's for the view State other values for other things and then you have the specific purposes right so the first one is a template Source directory slash account the second one is the account register ASP X so both of them quite predictable for an attacker right the third one looks

like pretty random uh generated like these 776 EC or whatever value that looks like arbitrary value however this is the same value that is used in your anti-cross height request forgery token um cookie right so the attacker is going to be able to read that and it's going to be able to um generate all these strings and therefore if they are able to leak this case then they will be able to generate those keys right so now the problem is you have to be very careful with not leaking the case uh the first way that the attacker can actually link the keys is if they find a file disclosure vulnerability in your application like for example an xxc vulnerability um also they may be some vulnerabilities in the framework like in 2010 and 15 that allow the attackers to actually leak those keys from Choice side um you may recognize this apparently you can actually leak the validation key and decryption key if you are not using custom errors at least in the past also I found like hundreds of validation and decryption keys in GitHub so if you are going to install something like a CMS or a block just by you know cloning the repo from GitHub and then installing that be very careful and always generate a new case never use the ones that are provided by this kind of projects same thing with one click installers do you get the latest version of this whatever CMS or whatever block application and then you install it and some of them will use a static keys that they won't generate per installation in those cases the attacker will be able to install the same CMS for him for example and then you know no leak the keys and then attack and get arbitrary call execution in your CMS block application or whatever so be very very careful with licking the key a good advice is to encrypt some sections of your web config file so you can actually ask IIS to encrypt this section so if you get like um XX vulnerability or a file disclosure vulnerability at least they won't be able to leave those keys so sign in is okay if the client doesn't need to modify the data so that's that's that's okay but what if this is a pure like client server that are sending like arbitrary messages then you cannot sign the the data because in that case the client needs to have the signature Keys the validation Keys which if the attacker is in control of the client it makes no sense so this in this case uh what you have to do is actually bind it right uh this is uh very simple it's very similar to the concept of the look ahead visualization in Java so basically right before instantiating those types that are controlled by the attacker just stop for a while and ask yourself if you want to really deserialize this uh disassembly name and this type name right so you're provided with this assembly name and type name and you can stop and apply a white list or a blacklist and then decide if you want to continue with the digitalization or stop right so this is done by implementing the serialization binary interface this is a good example by Jonathan Birch from Microsoft um basically just implement this interface and then prepare like a list of allowed types with the type names that you are going to allow and then a pre-loaded instance of those types this will be important I will show you a while later then just implement the override the bind to type and then check if the type that the attacker is sending is what is within the white list of allowed types and if that's the case then continue and return the pre-loaded type otherwise just throw an exception and avoid the um the the digitalization so then you have to actually instantiate the serialization binder assign it to the binder property and this realize right it's in so while I was reviewing some Microsoft product I found one of them being vulnerable to a remote execution through dc-alization and they had something they had a digitalization binder in place so it was like okay they are caring about this but then I

realized that the serialization binder was not that secure as they thought for example so this is not the actual Microsoft code but this is something that I put together in order to reproduce more or less what they were doing so I'm still in the process of disclosing this vulnerability I haven't been fixed so at some point what they were doing is they were checking if the name space starts with Microsoft dot put the name of a very popular Microsoft product here and if that's the case then load the type and return it otherwise return null so there are multiple vulnerabilities here first of all if you return null it may not do what you may think that is doing it's not going to avoid the digitalization it's going to fall back to the default binder which will bind anything which will bind including the remarkable execution Gadget type so it's not I have not protecting anything so the second problem here is that loading the type using reflection at this realization time is very costly and actually this will lead to the OS the attacker can easily de-occe your application because you are not using pre-loaded types if you remember the good example from from Jonathan he was actually preloading the types here and then just returning them in the method that gets executed during this realization and the third problem here is that this space this name space here is just huge it contains like thousands of applications of types right so even if the attacker cannot find a remarkable execution Gadget within this Microsoft dot something namespace they will be able probably chances are that they will be able to find what is known as a bypass Gadget and what is a bypass Gadget it's just a gadget so it starts with a digitalization callback and at some point instead of running arbitrary code for example or doing bad things they just run a nested digitalization operation the problem here is that even though you are protecting your outer deserialization operation the inner one the one that is run by this bypass Gadget has no binder in it right you have it's instantiated uh here and then just this realizing there is no binder assignment so that means that this this alization operation will let all the remote execution Gadget types to be processed and this realized and and the payloads to be executed so yeah this is basically some apps what I was talking about never return null because this will fall back to The Fault binder and that's bad also don't use um reflection to load types at runtime because that will deal as your application and the fourth approach is to replace it right so we saw the first one is stop using it if you don't need it the second one if the client doesn't need to change the data then sign it and verify that third approach is if this client needs to actually send you data then use a binder to control what types gets deserialized and the fourth approach is if the architecture of your application allows you to do that change these formats that include the type discriminators the type information change that with something else and a good alternative here is Google protocol buffers where you agree the client and the server agree in what they are going to exchange and then they exchange that's the data so if they are going to exchange an instance of outcome.acme.user then they will agree they will know in advance that they are going to exchange that and then they will just share the name of the user the last name of the user the address of the user but no type information no property type names or nothing like that and also if you don't really need the type information like we said before just don't use it and that will make those Json and XML serializers to be safe at least from this kind of attacks so with that um Mahalo and if you have any questions I think we have time right to take questions oh sorry um [Music] so that's that's this approach and and basically the

only cases where you really need to include type information that I can think of is if you are dealing with for example uh polymorphism types where you may send I don't know an instance of an animal but that may you need the information to need to to know if you need to reconstruct a dog or a cat for example so you need to send information the type information with that and that's the whole purpose of all this time name handling things uh digestion JavaScript as Clips serializer these simple type resolver and so on so those are meant to deal with some of the object-oriented programming features like polymorphism generics and so on so if you you should be only be sending things like dtos and poios right something very simple if you find yourself that you require things like polymorphism then maybe you need to rethink what you're sending Tim I don't know Astrid there you go I love this slide so I would just put it again so yeah this is the the whole design flow in digitalization is that they allow you to change the type discriminators and if that happens then attackers can arbitrary instantiate any type and that's the big problem here any questions all right so thank you foreign [Applause]